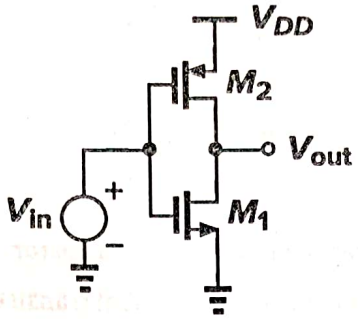
MID SEMESTER EXAMINATION (QP+SCHEME), ANALOG CIRCUITS (ECE 2126)

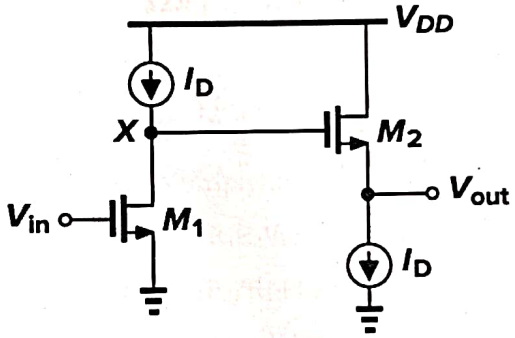
Type: MCQ

Q1. The voltage gain for the circuit shown below is _____. Consider $\lambda \neq 0$.(1)



1. $-(g_{m1} \times g_{m2})(r_{o1} || r_{o2})$
2. **$-(g_{m1}+g_{m2})(r_{o1} || r_{o2})$**
3. $-(g_{m1})(r_{o1} || r_{o2})$
4. $-(g_{m2})(r_{o1} || r_{o2})5$

Q2. Vout/Vin for the circuit shown below is _____. Consider $\lambda \neq 0$. (1)

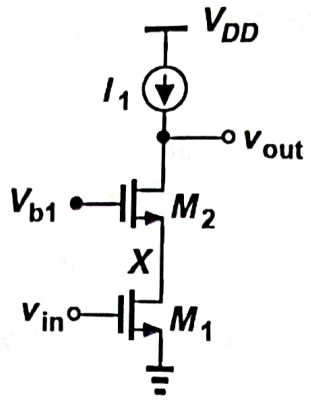


1. **$-g_{m1}r_{o1}g_{m2}r_{o2}/1+ g_{m2}r_{o2}$**
2. $g_{m1}r_{o1}$
3. $-g_{m1}r_{o1}$
4. $g_{m1}r_{o1}g_{m2}r_{o2}/1+ g_{m2}r_{o2}$

Q3. A MOSFET is biased at a drain current of 0.5 mA. If $\mu_n C_{ox} = 100\ \mu A/V^2$, $W/L = 10$, and $\lambda = 0.1\ V^{-1}$, intrinsic gain of the device is_____ (1)

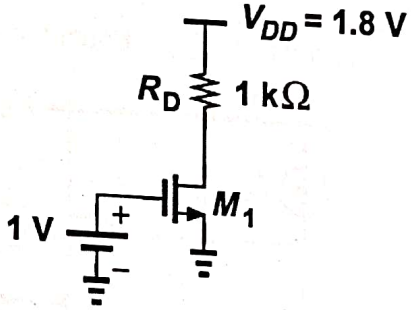
1. 10
2. **20**
3. 5
4. 15

Q4. Two transisitor M_1 and M_2 are connected as follows. The short circuit transconductance G_M of the circuit is_____ (1)



1. Sum of Transconductance of individual transistor
2. Product of Transconductance of individual transistor
3. **Equals to Transconductance of M_1**
4. Equals to Transconductance of M_2

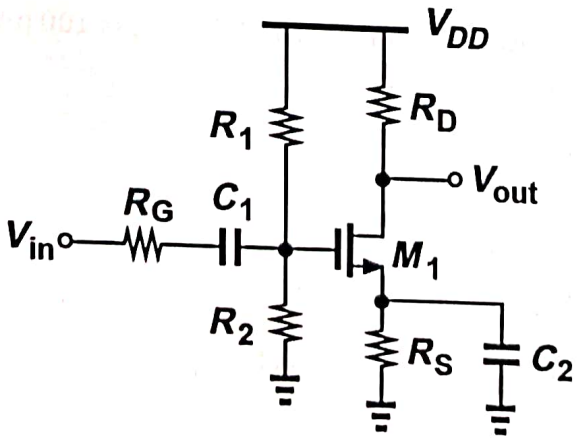
Q5. Assume $\lambda = 0$, $\mu_n C_{ox} = 100\ \mu A/V^2$, V_{TH}=0.4V. If the device operates at the edge of saturation the W/L of M_1 is _____. (1)



1. 33
2. 99
3. **66**
4. 11

Type: DES

Q6. Design the CS stage shown below. Consider the voltage gain of -5, an input impedance of 50KΩ, a power budget of 5 mW. Assume $\mu_n C_{ox} = 100\ \mu A/V^2$, $V_{TH} = 0.5$ V, $\lambda = 0$, and$V_{DD} = 1.8$ V. Also, assume a voltage drop of 400 mV across R_S, R_G=0, V_{GS}=V_{TH}+0.2, I_{DS}=2.77mA, $I_{R1,R2}$ = 0.01mA. (4)



Solution: {1 Mark for R_D, 1 Mark for R_S, 1 Mark for W/L and 1 Mark for R_1, R_2 }

$$\text{Power Budget} = V_{DD} I$$

$$I = 2.78\text{mA}$$

$$I_{DS} = 2.77\text{mA}, I_{R1,R2} = 0.01\text{mA}.$$

$$V_{GS} = V_{TH} + 0.2 = 0.7\text{V}$$

$$g_m = 2I_{DS}/V_{GS} - V_{TH} = 2.77 \times 10^{-2}$$

$$Av = g_m R_D$$

$$\mathbf{R_D = 180.5\Omega}$$

$$V_S = I_{DS} R_S$$

$$\mathbf{R_S = 144.404\Omega}$$

$$I_{DS} = 0.5\, \mu_n C_{ox} (W/L)(V_{GS} - V_{TH})^2$$

$$2.77 \times 10^{-3} = 0.5 \times 100 \times 10^{-6} (W/L)\, (0.7 - 0.5)^2$$

$$\mathbf{W/L = 1385}$$

$$V_{GS} = V_G - V_S$$

$$0.7 = V_G - 0.4$$

$$V_G = 1.1\text{V}$$

$$V_G = R_2 V_{DD} / R_1 + R_2$$

Multiplying by R_1

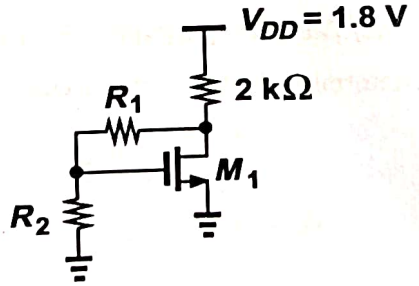
$$V_G R_1 = R_1 R_2 V_{DD} / R_1 + R_2$$

$$R_{IN} = R_1 R_2 / R_1 + R_2 = 50\text{K}\Omega$$

On solving

$$\mathbf{R_1 = 81.8\text{K}\Omega \text{ and } R_2 = 128.5\text{K}\Omega}$$

Q7. In the circuit shown, below the drain current (I_{DS1}) in MOSFET (M_1) is 0.5 mA. If W/L = 50/0.18, $\mu_n C_{ox} = 100\ \mu\text{A/V}^2$, $V_{TH} = 0.5$ V, $\lambda = 0$. Calculate the values of R_1 and R_2 such that these resistors carry a current equal to one-tenth of I_{DS1}. Consider that the M1 is operating in saturation. (3)



Solution: {1.5 Marks for R_1 and 1.5 Marks for R_2}

$$I_{R1,R2} = I_{DS1}/10 = 0.05\text{mA}$$

$$I_{DS1} = 0.5\, \mu_n C_{ox} (W/L)(V_{GS1} - V_{TH})^2$$

$$0.5\times10^{-3}=0.5\times100\times10^{-6}(50/0.18)\ (V_{GS1}-0.5)^2$$

$$V_{GS1}=0.689V$$

$$R_2=0.689/\ 0.05\times10^{-3}$$

$$\mathbf{R_2=13.78K\Omega}$$

$$1.8-\ 0.55\times10^{-3}\times2\times10^{3}\ =V_{D1}$$

$$V_{D1}=0.7V$$

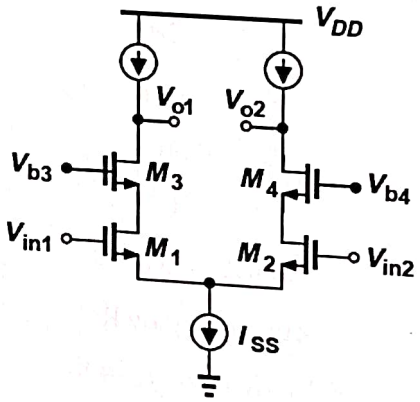
$$I_{R1}=V_{D1}-V_{GS1}/R_1$$

$$R_1=V_{D1}-V_{GS1}/\ I_{R1}$$

$$R_1=0.7-0.689/0.05\times10^{-3}$$

$$\mathbf{R_1=220\Omega}$$

Q8. Determine V_{O1}-V_{O2}/V_{in1}-V_{in2} for the circuit shown below. Consider the differential input is applied to the circuit, M_1-M_2, M_3-M_4 are matched, Current source are ideal, and $\lambda\neq0$. (3))



Solution: {1.5 Marks for single-ended gain, 1.5 Marks for double-ended gain}

Consider the half-circuit

$$\mathbf{V_{O1}/V_{ID}=-0.5g_{m1}\{r_{o1}+r_{o3}+g_{m3}r_{o3}r_{o1}\}}$$

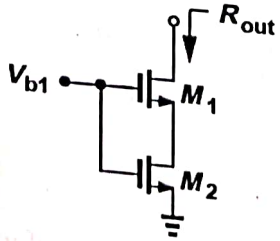
Since M_1-M_2, M_3-M_4 are matched

$$\mathbf{V_{O2}/V_{ID}=0.5g_{m1}\{r_{o1}+r_{o3}+g_{m3}r_{o3}r_{o1}\}}$$

$$V_{O1}-V_{O2}/V_{ID}=-g_{m1}\{r_{o1}+r_{o3}+g_{m3}r_{o3}r_{o1}\}$$

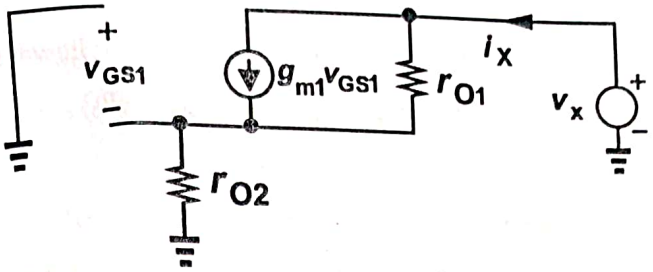
$$\mathbf{V_{O1}-V_{O2}/V_{in1}-V_{in2}=-g_{m1}\{r_{o1}+r_{o3}+g_{m3}r_{o3}r_{o1}\}}$$

Q9. Using a small signal model determine the R_{out} in the circuit shown. Consider $\lambda \neq 0$. (3)



Solution:{1.5 Marks for small signal model, 1.5 Marks for expression of R_{OUT}}

M_2 will offer a resistance of r_{O2}, replacing M_2 by r_{O2} and drawing the small signal model



Applying KVL

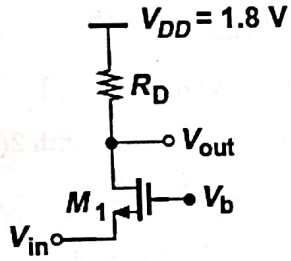
$V_X = (i_X - g_{m1}V_{GS1})r_{O1} + i_X r_{O2}$

Also, $V_{GS1} = -i_X r_{O2}$

On solving

$$R_{OUT} = V_X/i_X = r_{O2} + r_{O1} + g_{m1} r_{O1} r_{O2}$$

Q10. The circuit shown below must provide a voltage gain of 4 and an input impedance of 50. If $I_{DS} = 0.5$ mA, $\mu_n C_{ox} = 100\ \mu A/V^2$, and $\lambda = 0$, determine the values of R_D and W/L. (3)



Solution: {1 Mark for R_D and 1 Mark for W/L}

Input Impedance $= 1/g_m = 50$

$Av = g_m R_D$

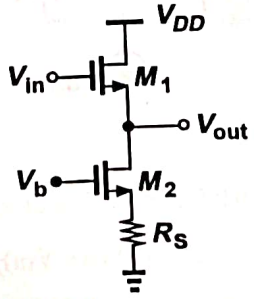
$$R_D = 200\Omega$$

$g_m = \{2\mu_n C_{ox}(W/L) I_{DS}\}^{0.5}$

$W/L = (g_m)^2/2\mu_n C_{ox} I_{DS} = (1/50)^2 / 2\times 100 \times 10^{-6} \times 0.5 \times 10^{-3}$

$$W/L = 4000$$

Q11. Compute the voltage gain of the circuit shown in the figure below. Consider $\lambda \neq 0$ for all the MOSFETS. (3)



Solution: {1.5 Marks for Z_{OUT} and 1.5 Marks for Av or 3 marks for correct gain expression if a student writes directly}

This is a common drain amplifier (source follower)

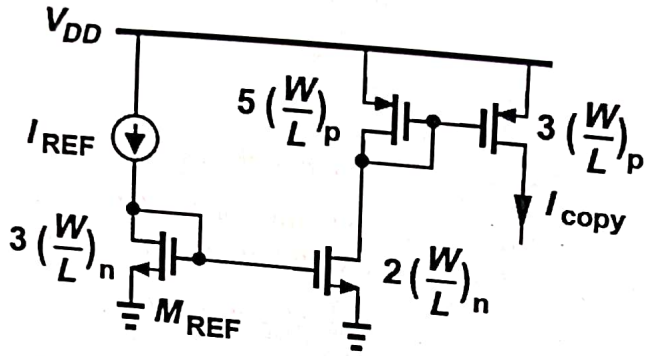
$$Zout = r_{o1} \,||\, 1/g_{m1} || (R_S + r_{o2} + g_{m2} r_{o2} R_S)$$

$$Av = g_{m1} Z_{out}$$

$$Av = g_{m1}\{ r_{o1} \,||\, 1/g_{m1} || (R_S + r_{o2} + g_{m2} r_{o2} R_S)\}$$

$$Av = \{r_{o1} \,||\, (R_S + r_{o2} + g_{m2} r_{o2} R_S)\}/[1/g_{m1} + \{r_{o1} \,||\, (R_S + r_{o2} + g_{m2} r_{o2} R_S)\}]$$

Q12. Calculate Icopy in the circuits shown if the I_{REF} =0.35mA. Assume all of the transistors operate in saturation. (2)



Solution:{1 Mark for I_X/I_{REF} and I_{COPY}/I_X, 1 Mark for I_{COPY} }

Let the current flowing in MOSFET with width $2(W/L)_n$ and $5(W/L)_p$ be I_X

$$I_X/I_{REF} = 2/3$$

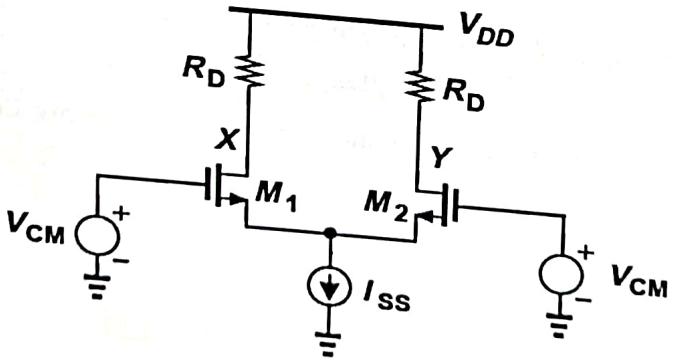
$$I_{COPY}/I_X = 3/5$$

Multiplying above equations

$$I_{COPY} = 2I_{REF}/5$$

$$I_{COPY} = 0.14mA$$

Q13. A MOS differential pair is driven with an input CM level of 1.6 V. If I_{SS} = 0.5 mA, V_{TH} = 0.5 V, and V_{DD} = 1.8 V, what is the maximum allowable load resistance? (2)



Solution: {1 Mark for the formula, 1 Mark for the value of $R_{D,max}$ }

$$R_{D,max} = 2(V_{DD} - V_{CM} + V_{TH})/I_{SS}$$

$$R_{D,max} = 2(1.8 - 1.6 + 0.5)/0.5 \times 10^{-3}$$

$$R_{D,max}=2.8K\Omega$$

Q14. A MOSFET is biased at a drain current of 0.5 mA. If $\mu_nC_{ox} = 100\ \mu A/V^2$, $W/L = 10$, and $\lambda = 0.1\ V^{-1}$, calculate its intrinsic gain. (2)

Solution: {1 Mark for gm, 0.5 Mark for ro and 0.5 Mark for intrinsic gain}

$$g_m=\{2\mu_nC_{ox}(W/L)I_{DS}\}^{0.5}$$

$$g_m=\{2\times100\times10^{-6}\times10\times0.5\times10^{-3}\}^{0.5}$$

$$\mathbf{g_m=10^{-3}\Omega^{-1}}$$

$$r_o=1/\lambda I_{DS}$$

$$r_o=1/0.1\times0.5\times10^{-3}$$

$$\mathbf{r_o=20K\Omega}$$

$$\mathbf{Intrinsic\ Gain=g_mr_o=20}$$